

IMURAN

Azathioprine

QUALITATIVE AND QUANTITATIVE COMPOSITION

Each tablet contains 50 mg of the active ingredient azathioprine.

PHARMACEUTICAL FORM

Azathioprine Tablets 50mg are yellow, round, biconvex, film-coated tablets which are scored, branded IM 5 and contains 50mg Azathioprine Ph. Eur. in each tablet.

CLINICAL PARTICULARS

THERAPEUTIC INDICATIONS

Azathioprine is used as an immunosuppressant antimetabolite either alone or, more commonly, in combination with other agents (usually corticosteroids) and procedures which influence the immune response. Therapeutic effect may be evident only after weeks or months and can include a steroid sparing effect, thereby reducing the toxicity associated with high dosage and prolonged usage of corticosteroids.

Azathioprine, in combination with corticosteroids and/or other immunosuppressive agents and procedures, is indicated to enhance the survival of organ transplants, such as renal transplants, cardiac transplants and hepatic transplants. It also reduces the corticosteroid requirements of renal transplant recipients.

Azathioprine, either alone or more usually in combination with corticosteroids and/or other drugs and procedures, has been used with clinical benefit (which may include reduction of dosage or discontinuation of corticosteroids) in a proportion of patients suffering from the following:

- severe rheumatoid arthritis
- systemic lupus erythematosus
- dermatomyositis and polymyositis
- auto-immune chronic active hepatitis
- pemphigus vulgaris
- polyarteritis nodosa
- auto-immune haemolytic anaemia
- chronic refractory idiopathic thrombocytopenic purpura

POSODOLOGY AND METHOD OF ADMINISTRATION

GENERAL

Azathioprine tablets should be administered at least 1 hour before or 3 hours after food or milk (*see Pharmacokinetic Properties*). The tablets should be swallowed whole.

POSODOLOGY POPULATIONS

- **Adults**

Transplants

Depending on the immunosuppressive regimen employed, a dosage of up to 5 mg/kg bodyweight/day may be given orally on the first day of therapy.

Maintenance dosage should range from 1 to 4 mg/kg bodyweight/day and must be adjusted according to clinical requirements and haematological tolerance.

Evidence indicates that azathioprine therapy should be maintained indefinitely, even if only low doses are necessary, because of the risk of graft rejection.

Other Indications

In general, starting dosage is from 1 to 3 mg/kg bodyweight/day, and should be adjusted, within these limits, depending on the clinical response (which may not be evident for weeks or months) and haematological tolerance.

When therapeutic response is evident, consideration should be given to reducing the maintenance dosage to the lowest level compatible with the maintenance of that response. If no improvement occurs in the patient's condition within three months, consideration should be given to withdrawing azathioprine.

The maintenance dosage required may range from less than 1 mg/kg bodyweight/day to 3 mg/kg bodyweight/day, depending on the clinical condition being treated and the individual patient response, including haematological tolerance.

- **Paediatric Population**

Transplants and Other Indications

See Posology and Method of Administration – Adults.

Overweight Paediatric Population

Children considered to be overweight may require doses at the higher end of the dose range and therefore close monitoring of response to treatment is recommended (see *Pharmacokinetic Properties*).

- **Elderly**

There is limited experience of the administration of azathioprine to elderly patients. Although the available data do not provide evidence that the incidence of side effects among elderly patients is higher than that among other patients treated with azathioprine, it is advisable to monitor renal and hepatic function, and to consider dosage reduction if there is impairment (see *Posology and Method of Administration - Renal Impairment and Hepatic Impairment*).

- **Renal Impairment**

In patients with renal insufficiency, consideration should be given to reducing the dosage (see *Special Warnings and Precautions for Use and Pharmacokinetic Properties*).

- **Hepatic Impairment**

In patients with hepatic insufficiency, consideration should be given to reducing the dosage (see *Special Warnings and Precautions for Use and Pharmacokinetic Properties*).

- **Drug Interactions**

When xanthine oxidase inhibitors, such as allopurinol, and azathioprine are administered concomitantly it is essential that only 25% of the usual dose of azathioprine is given since allopurinol decreases the rate of catabolism of azathioprine (see *Interaction with Other Medicinal Products and Other Forms of Interaction*).

- **TPMT-deficient Patients**

Patients with inherited little or no thiopurine S-methyltransferase (TPMT) activity are at increased risk for severe azathioprine toxicity from conventional doses of azathioprine and generally require substantial dose reduction. The optimal starting dose for homozygous deficient patients has not been established (see *Special Warnings and Precautions for Use*).

Most patients with heterozygous TPMT deficiency can tolerate recommended azathioprine doses, but some may require dose reduction. Genotypic and phenotypic tests of TPMT are available (see *Special Warnings and Precautions for Use*).

- **Patients with NUDT15 Variant**

Patients with inherited mutated NUDT15 gene are at increased risk for severe thiopurine toxicity, such as early leukopenia and alopecia, from conventional doses of thiopurine therapy and generally require substantial dose reduction.

Genotypic and phenotypic testing of NUDT15 variants should be considered before initiating thiopurine therapy in all patients (including paediatric patients) to reduce the risk of thiopurine-related severe leukocytopenia and alopecia, especially in Asian populations.

CONTRAINDICATIONS

Azathioprine is contraindicated in patients known to be hypersensitive to azathioprine or to any of the excipients listed in *List of Excipients*. Hypersensitivity to 6-mercaptopurine (6-MP) should alert the prescriber to probable hypersensitivity to azathioprine.

SPECIAL WARNINGS AND PRECAUTIONS FOR USE

Immunisation using a live organism vaccine has the potential to cause infection in immunocompromised hosts. Therefore, it is recommended that patients do not receive live

organism vaccines until at least three months after the end of their treatment with azathioprine (see *Interaction with Other Medicinal Products and Other Forms of Interaction*).

Co-administration of ribavirin and azathioprine is not advised. Ribavirin may reduce efficacy and increase toxicity of azathioprine (see *Interaction with Other Medicinal Products and Other Forms of Interaction*).

Monitoring

There are potential hazards in the use of azathioprine. It should be prescribed only if the patient can be adequately monitored for toxic effects throughout the duration of therapy.

Particular care should be taken to monitor haematological response and to reduce the maintenance dosage to the minimum required for clinical response.

It is suggested that during the first eight weeks of therapy, complete blood counts, including platelets, should be performed weekly or more frequently if high dosage is used or if severe renal and/or hepatic disorder is present. The blood count frequency may be reduced later in therapy, but it is suggested that complete blood counts are repeated monthly, or at least at intervals of not longer than three months.

At the first signs of an abnormal fall in blood counts, treatment should be interrupted immediately as leucocytes and platelets may continue to fall after treatment is stopped.

Patients receiving azathioprine should be instructed to report immediately any evidence of infection, unexpected bruising or bleeding or other manifestations of bone marrow depression. Bone marrow suppression is reversible if azathioprine is withdrawn early enough.

Azathioprine is hepatotoxic and liver function tests should be routinely monitored during treatment. More frequent monitoring may be advisable in those with pre-existing liver disease or receiving other potentially hepatotoxic therapy. The patient should be instructed to discontinue azathioprine immediately if jaundice becomes apparent.

Cholestasis of pregnancy has occasionally been reported in association with azathioprine therapy (see *Fertility, Pregnancy and Lactation*). Monitoring of 6-methyl mercaptopurine (6-MMP) should be considered in the presence of pruritus with elevated maternal total serum bile acids levels in second trimester of pregnancy to establish early diagnosis and minimise impact on the foetus. If cholestasis of pregnancy occurs, case by case assessment is necessary considering the risk-benefit profile of the product (potential withdrawal/dose reduction).

Thiopurine methyltransferase (TPMT)

There are individuals with an inherited deficiency of the enzyme thiopurine

methyltransferase (TPMT) who may be unusually sensitive to the myelosuppressive effect of azathioprine and prone to developing rapid bone marrow depression following the initiation of treatment with azathioprine. This problem could be exacerbated by coadministration with drugs that inhibit TPMT, such as olsalazine, mesalazine or sulphasalazine. Also a possible association between decreased TPMT activity and secondary leukaemias and myelodysplasia has been reported in individuals receiving 6-mercaptopurine (the active metabolite of azathioprine) in combination with other cytotoxics (see *Undesirable Effects*). Some laboratories offer testing for TPMT deficiency, although these tests have not been shown to identify all patients at risk of severe toxicity. Therefore close monitoring of blood counts is still necessary. The dosage of azathioprine may need to be reduced when this agent is combined with other drugs whose primary or secondary toxicity is myelosuppression (see *Interaction with Other Medicinal Products and Other Forms of Interaction*).

NUDT15 Mutation

Patients with inherited mutated NUDT15 gene are at increased risk for severe azathioprine toxicity, such as early leukopenia and alopecia, from conventional doses of thiopurine therapy. They generally require dose reduction, particularly those being NUDT15 variant homozygotes (see *Posology and Method of Administration*). The frequency of NUDT15 c.415C>T has an ethnic variability of approximately 10 % in East Asians, 4 % in Hispanics, 0.2 % in Europeans and 0 % in Africans. In any case, close monitoring of blood counts is necessary.

Hypersensitivity

Patients suspected to have previously presented a hypersensitivity reaction to 6-mercaptopurine should not be recommended to use its pro-drug azathioprine, and vice-versa, unless the patient has been confirmed as hypersensitive to the culprit drug with allergological tests, and tested negative for the other.

Renal and/or Hepatic Impairment

Caution is advised during the administration of azathioprine in patients with renal impairment and/or hepatic impairment. Consideration should be given to reducing the dosage in these patients and haematological response should be carefully monitored (see *Posology and Method of Administration and Pharmacokinetic Properties*)

Lesch-Nyhan Syndrome

Limited evidence suggests that azathioprine is not beneficial to patients with hypoxanthine-guanine-phosphoribosyltransferase deficiency (Lesch-Nyhan syndrome). Therefore, given the abnormal metabolism in these patients, it is not prudent to recommend that these patients should receive azathioprine.

Mutagenicity

Chromosomal abnormalities have been demonstrated in both male and female patients treated with azathioprine. It is difficult to assess the role of azathioprine in the development of these abnormalities.

Chromosomal abnormalities, which disappear with time, have been demonstrated in lymphocytes from the off-spring of patients treated with azathioprine. Except in extremely rare cases, no overt physical evidence of abnormality has been observed in the offspring of patients treated with azathioprine.

Azathioprine and long-wave ultraviolet light have been shown to have a synergistic clastogenic effect in patients treated with azathioprine for a range of disorders.

Effects on Fertility

Relief of chronic renal insufficiency by renal transplantation involving the administration of azathioprine has been accompanied by improved fertility in both male and female transplant recipients.

Carcinogenicity (see *Undesirable Effects*)

Patients receiving immunosuppressive therapy, including azathioprine are at an increased risk of developing lymphoproliferative disorders and other malignancies, notably skin cancers (melanoma and non-melanoma), sarcomas (Kaposi's and non-Kaposi's) and uterine cervical cancer *in situ*. The increased risk appears to be related to the degree and duration of immunosuppression. It has been reported that discontinuation of immunosuppression may provide partial regression of lymphoproliferative disorder.

A treatment regimen containing multiple immunosuppressants (including thiopurines) should therefore be used with caution as this could lead to lymphoproliferative disorders, some with reported fatalities. A combination of multiple immunosuppressants, given concomitantly increases the risk of Epstein-Barr virus (EBV)-associated lymphoproliferative disorders.

Reports of hepatosplenic T-cell lymphoma have been received when azathioprine is used alone or in combination with anti-TNF agents or other immunosuppressants. Although most reported cases occurred in the IBD population, there have also been cases reported outside of this population.

Patients receiving multiple immunosuppressive agents may be at risk of over-immunosuppression, therefore such therapy should be maintained at the lowest effective level. As is usual for patients with increased risk for skin cancer, exposure to sunlight and UV light should be limited, and patients should wear protective clothing and use

sunscreen with a high protection factor.

Macrophage Activation Syndrome

Macrophage Activation Syndrome (MAS) is a known, life-threatening disorder that may develop in patients with autoimmune conditions, in particular with inflammatory bowel disease (IBD), and there could potentially be an increased susceptibility for developing the condition with the use of azathioprine. If MAS occurs, or is suspected, evaluation and treatment should be started as early as possible, and treatment with azathioprine should be discontinued. Physicians should be attentive to symptoms of infection such as EBV and cytomegalovirus (CMV), as these are known triggers for MAS.

Metabolism and Nutrition Disorders

Administration of purine analogues (azathioprine and mercaptopurine) may interfere with the niacin pathway, potentially leading to nicotinic acid deficiency/pellagra. Few cases have been reported with the use of azathioprine, especially in patients with IBD (Crohn's disease, colitis ulcerative). Diagnosis of pellagra should be considered in a patient presenting with localised pigmented rash (dermatitis); gastroenteritis (diarrhoea); and widespread neurologic deficits, including cognitive decline (dementia). Dose reduction or discontinuation of azathioprine treatment may not be required if appropriate medical care with niacin/nicotinamide supplementation is initiated. However, careful benefit-risk assessment is required on a case-by-case basis.

Varicella Zoster Virus Infection (*see Undesirable Effects*)

Infection with varicella zoster virus (VZV; chickenpox and herpes zoster) may become severe during the administration of immunosuppressants. Caution should be exercised especially with respect to the following:

Before starting the administration of immunosuppressants, the prescriber should check to see if the patient has a history of VZV. Serologic testing may be useful in determining previous exposure. Patients who have no history of exposure should avoid contact with individuals with chickenpox or herpes zoster. If the patient is exposed to VZV, special care must be taken to avoid patients developing chickenpox or herpes zoster, and passive immunisation with varicella zoster immunoglobulin (VZIG) may be considered.

If the patient is infected with VZV, appropriate measures should be taken, which may include antiviral therapy and supportive care.

Progressive Multifocal Leukoencephalopathy (PML)

PML, an opportunistic infection caused by the JC virus, has been reported in patients receiving azathioprine with other immunosuppressive agents. Immunosuppressive therapy should be withheld at the first sign or symptoms suggestive of PML and appropriate evaluation undertaken to establish a diagnosis (*see Undesirable Effects*).

Hepatitis B (See Undesirable Effects)

Hepatitis B carriers (defined as patients positive for hepatitis B surface antigen [HBsAg] for more than six months), or patients with documented past HBV infection, who receive immunosuppressive drugs are at risk of reactivation of HBV replication, with asymptomatic increases in serum HBV DNA and ALT levels. Local guidelines may be considered including prophylactic therapy with oral anti-HBV agents.

Xanthine oxidase inhibitors

If allopurinol, oxipurinol and/or thiopurinol are given concomitantly with azathioprine, the dosage of azathioprine must be reduced to a quarter of the original dose (see *Posology and Method of Administration*).

Neuromuscular agents

Special care is necessary when azathioprine is given concomitantly with neuromuscular acting agents like tubocurarine or succinylcholine. It can also potentiate the neuromuscular block that is produced by depolarising agents such as succinylcholine (see *Interaction with Other Medicinal Products and Other Forms of Interactions*). Patients should be advised to inform their anaesthesiologist of their treatment with azathioprine prior to surgery.

INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION**Vaccines**

The immunosuppressive activity of azathioprine could result in an atypical and potentially deleterious response to live vaccines. It is therefore recommended that patients do not receive live vaccines until at least three months after the end of their treatment with azathioprine. (see *Special Warnings and Precautions for Use*).

A diminished response to killed vaccines is likely and such a response to hepatitis B vaccine has been observed among patients treated with a combination of azathioprine and corticosteroids.

A small clinical study has indicated that standard therapeutic doses of azathioprine do not deleteriously affect the response to polyvalent pneumococcal vaccine, as assessed on the basis of mean anti-capsular specific antibody concentration.

Effect of concomitant drugs on azathioprine**Ribavirin**

Ribavirin inhibits the enzyme inosine monophosphate dehydrogenase (IMPDH), leading to a lower production of the active 6-thioguanine nucleotides. Severe myelosuppression has been reported following concomitant administration of azathioprine and ribavirin; therefore co-administration is not advised (see *Special Warnings and Precautions for Use*).

Cytostatic/myelosuppressive agents (see *Special Warnings and Precautions for Use*)

Where possible, concomitant administration of cytostatic drugs, or drugs which may have a myelosuppressive effect, such as penicillamine, should be avoided. There are conflicting clinical reports of interactions, resulting in serious haematological abnormalities, between azathioprine and trimethoprim/sulfamethoxazole.

There have been case reports suggesting that haematological abnormalities may develop due to the concomitant administration of azathioprine and ACE inhibitors.

It has been suggested that cimetidine and indomethacin may have myelosuppressive effects which may be enhanced by concomitant administration of azathioprine.

Allopurinol/oxipurinol/thiopurinol and other xanthine oxidase inhibitors

Xanthine oxidase activity is inhibited by allopurinol, oxipurinol and thiopurinol, which results in reduced conversion of biologically active 6-thioinosinic acid to biologically inactive 6-thiouric acid. When allopurinol, oxipurinol and/or thiopurinol are given concomitantly with 6-mercaptopurine or azathioprine, the dose of 6-mercaptopurine and azathioprine should be reduced to 25% of the original dose (see *Posology and Method of Administration*).

Other xanthine oxidase inhibitors, such as febuxostat may decrease the metabolism of azathioprine. Concomitant administration is not recommended as data are insufficient to determine an adequate dose reduction.

Aminosalicylates

There is *in vitro* and *in vivo* evidence that aminosalicylate derivatives (e.g. olsalazine, mesalazine or sulphasalazine) inhibit the TPMT enzyme. Therefore, lower doses of azathioprine may need to be considered when administered concomitantly with aminosalicylate derivatives (see *Special Warnings and Precautions for Use*).

Methotrexate

Methotrexate (20 mg/m² orally) increased 6-mercaptopurine AUC by approximately 31% and methotrexate (2 or 5 g/m² intravenously) increased 6-mercaptopurine AUC by 69% and 93%, respectively.

Infliximab

An interaction has been observed between azathioprine and infliximab. Patients receiving ongoing azathioprine experienced transient increases in 6-TGN (6-thioguanine nucleotide, an active metabolite of azathioprine) levels and a decrease in the mean leukocyte count in the initial weeks following infliximab infusion, which returned to previous levels after three months.

Neuromuscular agents

There is clinical evidence that azathioprine antagonizes the effect of non-depolarising muscle relaxants such as curare, d-tubocurarine and pancuronium. Experimental data confirm that azathioprine reverses the neuromuscular blockade produced by d-tubocurarine and show that azathioprine potentiates the neuromuscular blockade produced by succinylcholine (see *Special Warnings and Precautions for Use*).

Effect of azathioprine on other drugs

Anticoagulants

Inhibition of the anticoagulant effect of warfarin and acenocoumarol has been reported when co-administered with azathioprine; therefore higher doses of the anticoagulant may be needed. It is recommended that coagulation tests are closely monitored when anticoagulants are concurrently administered with azathioprine.

FERTILITY, PREGNANCY AND LACTATION

Pregnancy

Substantial transplacental and transamniotic transmission of azathioprine and its metabolites from the mother to the foetus have been shown to occur.

Azathioprine should not be given to patients who are pregnant or likely to become pregnant in the near future without careful assessment of risk versus benefit.

Evidence of the teratogenicity of azathioprine in man is equivocal. As with all cytotoxic chemotherapy, adequate contraceptive precautions should be advised when either partner is receiving azathioprine.

There have been reports of premature birth and low birth weight following maternal exposure to azathioprine, particularly in combination with corticosteroids. There have also been reports of spontaneous abortion following either maternal or paternal exposure.

Leukopenia and/or thrombocytopenia have been reported in a proportion of neonates whose mothers took azathioprine throughout their pregnancies. Extra care in haematological monitoring is advised during pregnancy.

Cholestasis of pregnancy has occasionally been reported in association with azathioprine therapy. Early diagnosis and discontinuation of azathioprine may minimise impact on the foetus. However, a careful assessment of benefit to the mother and impact on the foetus should be performed, if cholestasis of pregnancy is confirmed (see *Special Warnings and Precautions for Use*).

Breastfeeding

6-Mercaptopurine has been identified in the colostrum and breast-milk of women receiving azathioprine treatment. It is recommended that mothers receiving azathioprine should not breastfeed.

Fertility

The specific effect of azathioprine therapy on human fertility is unknown.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

There are no data on the effect of azathioprine on driving performance or the ability to operate machinery. A detrimental effect on these activities cannot be predicted from the pharmacology of the drug.

UNDESIRABLE EFFECTS

Summary of the safety profile

For azathioprine, there is no modern clinical documentation which can be used as support for determining the frequency of undesirable effects. Undesirable effects may vary in their incidence depending on the indication.

Tabulated summary of adverse reactions

The following convention has been utilised for the classification of frequency: Very common $\geq 1/10$; common $\geq 1/100$ and $< 1/10$; uncommon $\geq 1/1000$ and $< 1/100$; rare $\geq 1/10,000$ and $< 1/1000$; very rare $< 1/10,000$ and not known (cannot be estimated from the available data).

System Organ Class	Frequency	Adverse reactions
Infections and infestations	Very common	viral, fungal and bacterial infections in transplant patients receiving azathioprine in combination with other immunosuppressants
	Uncommon	viral, fungal and bacterial infections in other patient populations
	Very Rare	cases of JC virus associated PML have been reported following the use of azathioprine in combination with other immunosuppressants (see <i>Special Warnings and Precautions for Use</i>)
Neoplasms benign, malignant and unspecified (including cysts and polyps)	Rare	neoplasms including lymphoproliferative disorders, skin cancers (melanoma and non- melanoma), sarcomas (Kaposi's and non- Kaposi's) and uterine cervical cancer <i>in situ</i> , acute myeloid leukaemia and myelodysplastic syndrome (see <i>Special Warning and Precautions for Use</i>)
	Not known	Hepatosplenic T-cell lymphoma (see <i>Special Warnings and Precautions for Use</i>)

Blood and lymphatic system disorders	Very common	leukopenia, bone marrow depression
	Common	thrombocytopenia
	Uncommon	anaemia
	Rare	agranulocytosis, pancytopenia, aplastic anaemia, anaemia megaloblastic
Immune system disorders	Uncommon	hypersensitivity
	Very rare	Stevens-Johnson syndrome and toxic epidermal necrolysis
Respiratory, thoracic and mediastinal disorders	Very rare	reversible pneumonitis
Gastrointestinal disorders	Common	nausea
	Uncommon	pancreatitis
	Very rare	colitis, diverticulitis and intestinal perforation reported in transplant population, severe diarrhoea in inflammatory bowel disease population
Hepatobiliary disorders	Uncommon	cholestasis
	Rare	life-threatening liver injury
Skin and subcutaneous tissue disorders	Rare	alopecia
	Not known	acute febrile neutrophilic dermatosis (Sweet's syndrome), photosensitivity reaction
Investigations	Uncommon	liver function test abnormal

Description of selected adverse reactions

Infections and Infestations

Patients receiving azathioprine alone or in combination with other immunosuppressants, particularly corticosteroids, have shown increased susceptibility to viral, fungal and bacterial infections, including severe or atypical infection, and reactivation with VZV, hepatitis B and other infectious agents.

Neoplasms Benign, Malignant and Unspecified (including cysts and polyps)

Patients receiving immunosuppressive therapy, including azathioprine are at an increased risk of developing lymphoproliferative disorders and other malignancies, notably skin

cancers (melanoma and non-melanoma), sarcomas (Kaposi's and non-Kaposi's) and uterine cervical cancer *in situ*. The increased risk appears to be related to the degree and duration of immunosuppression. It has been reported that discontinuation of immunosuppression may provide partial regression of the lymphoproliferative disorder.

There have been rare reports of acute myeloid leukaemia and myelodysplasia (some in association with chromosomal abnormalities).

Blood and Lymphatic System Disorders

Azathioprine may be associated with a dose-related, generally reversible, depression of bone marrow function, most frequently expressed as leukopenia, but also sometimes as anaemia and thrombocytopenia and rarely as agranulocytosis, pancytopenia and aplastic anaemia. These occur particularly in patients predisposed to myelotoxicity, such as those with TPMT deficiency and renal or hepatic insufficiency and in patients failing to reduce the dose of azathioprine when receiving concurrent allopurinol therapy.

Reversible, dose-related increases in mean corpuscular volume and red cell haemoglobin content have occurred in association with azathioprine therapy. Megaloblastic bone marrow changes have also been observed but severe megaloblastic anaemia and erythroid hypoplasia are rare.

Immune System Disorders

Several different clinical syndromes, which appear to be idiosyncratic manifestations of hypersensitivity, have been described occasionally following administration of azathioprine. Clinical features include general malaise, dizziness, nausea, vomiting, diarrhoea, fever, rigors, exanthema, rash, vasculitis, erythema nodosum, myalgia, arthralgia, hypotension, renal dysfunction, hepatic dysfunction and cholestasis (see *Undesirable Effects – Hepatobiliary disorders*).

In many cases, rechallenge has confirmed an association with azathioprine.

Immediate withdrawal of azathioprine and institution of circulatory support where appropriate have led to recovery in the majority of cases.

Other marked underlying pathology has contributed to the very rare deaths reported.

Following a hypersensitivity reaction to azathioprine, the necessity for continued administration of azathioprine should be carefully considered on an individual basis.

Gastrointestinal Disorders

Some patients experience nausea when first given oral azathioprine. This appears to be relieved by administering the tablets after meals. However, administration of azathioprine tablets after meals may reduce oral absorption, therefore monitoring for therapeutic efficacy

should be considered after administration in this way (*see Pharmacokinetic Properties*).

Serious complications, including colitis, diverticulitis and bowel perforation, have been described in transplant recipients receiving immunosuppressive therapy. However, the aetiology is not clearly established and high-dose corticosteroids may be implicated. Severe diarrhoea, recurring on rechallenge, has been reported in patients treated with azathioprine for inflammatory bowel disease. The possibility that exacerbation of symptoms might be drug-related should be borne in mind when treating such patients.

Pancreatitis has been reported in a small percentage of patients on azathioprine therapy, particularly in renal transplant patients and those diagnosed as having inflammatory bowel disease. There are difficulties in relating the pancreatitis to the administration of one particular drug, although rechallenge has confirmed an association with azathioprine on occasions.

Hepatobiliary Disorders

Cholestasis and deterioration of liver function have occasionally been reported in association with azathioprine therapy and are usually reversible on withdrawal of therapy. This may be associated with symptoms of a hypersensitivity reaction (*see Undesirable Effects - Immune system disorders*).

Rare, but life-threatening hepatic damage associated with chronic administration of azathioprine has been described primarily in transplant patients. Histological findings include sinusoidal dilatation, peliosis hepatis, venoocclusive disease and nodular regenerative hyperplasia. In some cases withdrawal of azathioprine has resulted in either a temporary or permanent improvement in liver histology and symptoms.

Skin and Subcutaneous Tissue Disorders

Hair loss has been described on a number of occasions in patients receiving azathioprine and other immunosuppressive agents. In many instances the condition resolved spontaneously despite continuing therapy. The relationship between alopecia and azathioprine treatment is uncertain.

OVERDOSE

Symptoms

Unexplained infection, ulceration of the throat, bruising and bleeding are the main signs of overdosage with azathioprine and result from bone marrow depression, which may be maximal after 9 to 14 days. These signs are more likely to be manifested following chronic overdosage, rather than after a single acute overdose. There has been a report of a patient who ingested a single overdose of 7.5 g of azathioprine. The immediate toxic effects of this overdose were nausea, vomiting and diarrhoea, followed by mild leukopenia and mild abnormalities in liver function. Recovery was uneventful.

Management

As there is no specific antidote, blood counts should be closely monitored and general supportive measures, together with appropriate blood transfusion, instituted if necessary. Active measures (such as the use of activated charcoal) may not be effective in the event of azathioprine overdose unless the procedure can be undertaken within 60 minutes of ingestion.

Further management should be as clinically indicated or as recommended by the national poisons centre, where available. The value of dialysis in patients who have taken an overdose of azathioprine is not known, though azathioprine is partially dialysable.

PHARMACOLOGICAL PROPERTIES

PHARMACODYNAMIC PROPERTIES

Pharmacotherapeutic group: ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS – Immunosuppressants – Other immunosuppressants

ATC code: L04AX01

Mechanism of Action

Azathioprine is a pro-drug of 6-mercaptopurine (6-MP). 6-MP is inactive but acts as a purine antagonist and requires cellular uptake and intracellular anabolism to thioguanine nucleotides (TGNs) for immunosuppression. The TGNs and other metabolites (e.g. 6-methyl- mercaptopurine ribonucleotides) inhibit *de novo* purine synthesis and purine nucleotide interconversions. The TGNs are also incorporated into nucleic acids and this contributes to the immunosuppressive effects of the drug. Other potential mechanisms of azathioprine include the inhibition of many pathways in nucleic acid biosynthesis, hence preventing proliferation of cells involved in determination and amplification of the immune response.

Because of these mechanisms, the therapeutic effect of azathioprine may be evident only after several weeks or months of treatment.

The activity of the methylnitroimidazole moiety, a metabolite of azathioprine but not 6-MP, has not been defined clearly. However, in several systems it appears to modify the activity of azathioprine as compared with that of 6-MP.

Pharmacodynamic Effects

Plasma levels of azathioprine and 6-MP do not correlate well with the therapeutic efficacy or toxicity of azathioprine, and therefore have no prognostic value.

PHARMACOKINETIC PROPERTIES

Absorption

The absorption of azathioprine is incomplete and variable. The median (range) absolute bioavailability of 6-MP after administration of azathioprine 50 mg is 47% (27 – 80%). The extent of absorption of azathioprine is similar across the gastrointestinal tract, including the stomach, jejunum, and cecum. However, the extent of 6-MP absorption, after azathioprine administration is variable and differs between the sites of absorption, with the highest extent of absorption in the jejunum, followed by the stomach and then the cecum.

Although there are no food effect studies with azathioprine, pharmacokinetic studies with 6-MP have been conducted that are relevant to azathioprine. The mean relative bioavailability of 6-MP was approximately 26% lower following administration with food and milk compared to an overnight fast. 6-MP is not stable in milk due to the presence of xanthine oxidase (30% degradation within 30 minutes) (see *Pharmacokinetics: Biotransformation*). Azathioprine should be administered at least 1 hour before or 3 hours after food or milk (see *Dosology and Method of Administration*).

Distribution

The volume of distribution at steady state (V_{dss}) of azathioprine is unknown. The mean ($\pm$ SD) apparent V_{dss} of 6-MP is 0.9 (± 0.8) L/kg, although this may be an underestimate because 6-MP is cleared throughout the body (and not just in the liver).

Concentrations of 6-MP in cerebrospinal fluid (CSF) are low or negligible after IV or oral administration of 6-MP.

Biotransformation

Azathioprine is rapidly broken down *in vivo* by glutathione-S-transferase into 6-MP and a methylnitroimidazole moiety. The 6-MP readily crosses cell membranes and is extensively metabolized by many multi-step pathways to active and inactive metabolites, with no one enzyme predominating. Because of the complex metabolism, inhibition of one enzyme does not explain all cases of lack of efficacy and/or pronounced myelosuppression. The predominant enzymes responsible for the metabolism of 6-MP or its downstream metabolites are: the polymorphic enzyme thiopurine S-methyltransferase (TPMT) (see *Special Warnings and Precautions for Use*), xanthine oxidase (see *Interactions with Other Medicinal Products and Other Forms of Interaction*), inosine monophosphate dehydrogenase (IMPDH) (see *Interactions with Other Medicinal Products and Other Forms of Interaction*), and hypoxanthine guanine phosphoribosyltransferase (HPRT). Additional enzymes involved in the formation of active and inactive metabolites are: guanosine monophosphate synthetase (GMPS, which form TGNs) and inosine triphosphate pyrophosphatase (ITPase). Azathioprine itself is also metabolized by aldehyde oxidase to form 8-hydroxy azathioprine, which may be active. There are also multiple inactive metabolites formed via other pathways.

There is evidence that polymorphisms in the genes encoding the different enzyme systems involved with metabolism of azathioprine may predict adverse drug reactions to azathioprine therapy.

Thiopurine S-Methyl Transferase (TPMT)

TPMT activity is inversely related to red blood cell 6-MP derived thioguanine nucleotide concentration, with higher thioguanine nucleotide concentrations resulting in greater reductions in white blood cell and neutrophil counts. Individuals with TPMT deficiency develop very high cytotoxic thioguanine nucleotide concentrations.

Genotypic testing can determine the allelic pattern of a patient. Currently, 3 alleles—TPMT*2, TPMT*3A and TPMT*3C—account for about 95% of individuals with reduced levels of TPMT activity. Approximately 0.3% (1:300) of patients have two non-functional alleles (homozygous- deficient) of the TPMT gene and have little or no detectable enzyme activity. Approximately 10% of patients have one TPMT non- functional allele (heterozygous) leading to low or intermediate TPMT activity and 90% of individuals have normal TPMT activity with two functional alleles. There may also be a group of approximately 2% who have very high TPMT activity. Phenotypic testing determines the level of thiopurine nucleotides or TPMT activity in red blood cells and can also be informative (*see Special Warnings and Precautions for Use*).

Elimination

After oral administration of 100mg 35S-azathioprine, 50% of the radioactivity was excreted in the urine and 12% in the faeces after 24 hours. In the urine, the major compound was the inactive oxidised metabolite, thiouric acid. Less than 2% was excreted in the urine as azathioprine or 6-MP. Azathioprine has a high extraction ratio with a total clearance greater than 3L/min in normal volunteers. There is no data on the renal clearance or half-life of azathioprine. The renal clearance of 6-MP and the half-life of 6-MP are 191 mL/min/m² and 0.9 hour respectively.

Special Patient Populations

Elderly

No specific studies have been carried out in the elderly (*see Posology and Method of Administration*).

Overweight Paediatric Population

In a US clinical study, 18 children (aged 3 to 14 years) were evenly divided into two groups; either a weight to height ratio above or below the 75th percentile. Each child was on maintenance treatment of 6-MP and the dosage was calculated based on their body surface area. The mean AUC (0- ∞) of 6-MP in the group above the 75th percentile was 2.4 times lower than that for the group below the 75th percentile. Therefore, children considered to be overweight may require azathioprine doses at the higher end of the dose

range and close monitoring of response to treatment is recommended (see *Posology and Method of Administration*)

Renal Impairment

Studies with azathioprine have shown no difference in 6-MP pharmacokinetics in uremic patients compared to renal transplant patients. Since little is known about the active metabolites of azathioprine in renal impairment, consideration should be given to reducing the dosage in patients with impaired renal function (see *Posology and Method of Administration*).

Azathioprine and/or its metabolites are eliminated by haemodialysis, with approximately 45% of radioactive metabolites eliminated during dialysis of eight hours.

Hepatic Impairment

A study with azathioprine was performed in three groups of renal transplant patients: those without liver disease, those with hepatic impairment (but no cirrhosis) and those with hepatic impairment and cirrhosis. The study demonstrated that 6-mercaptopurine exposure was 1.6 times higher in patients with hepatic impairment (but no cirrhosis) and six times higher in patients with hepatic impairment and cirrhosis, compared to patients without liver disease. Therefore, consideration should be given to reducing the dosage in patients with impaired hepatic function (see *Posology and Method of Administration*).

PRE-CLINICAL SAFETY DATA

Reproductive toxicology

Studies in pregnant rats, mice and rabbits using azathioprine in dosages from 5 to 15 mg/kg bodyweight/ day over the period of organogenesis have shown varying degrees of foetal abnormalities.

Teratogenicity was evident in rabbits at 10 mg/kg bodyweight/day.

PHARMACEUTICAL PARTICULARS

List of Excipients

Tablets:

Lactose monohydrate, maize starch, pregelatinised starch, magnesium stearate, stearic acid, hypromellose, macrogol 400, purified water

Incompatibilities

None reported for azathioprine tablets.

Shelf-Life

The expiry date is indicated on the packaging.

Special Precautions for Storage

Store below 30°C. Protect from light.

Nature and Contents of Container**Tablets:**

Packed in foil blister strips of 4 x 25's.

Special Precautions for Disposal and Other Handling**Tablets:****Safe handling**

Health professionals who handle uncoated azathioprine tablets should follow guidelines for the handling of cytotoxic drugs according to prevailing local recommendations and/ or regulations.

Provided that the film-coating is intact, there is no risk in handling film-coated azathioprine tablets. Film-coated azathioprine tablets should not be divided and, provided the coating is intact, no additional precautions are required when handling them.

Disposal

Azathioprine tablets should be disposed of in a manner appropriate to the prevailing local regulatory requirements for the destruction of dangerous substances.

Version number of package insert: CCDSv27/01/0724

Date of revision of package insert: 01 Jul 2024

**MANUFACTURER:**

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